National 5 Engineering Science

Yenka Practical Tasks – Analogue Electronics

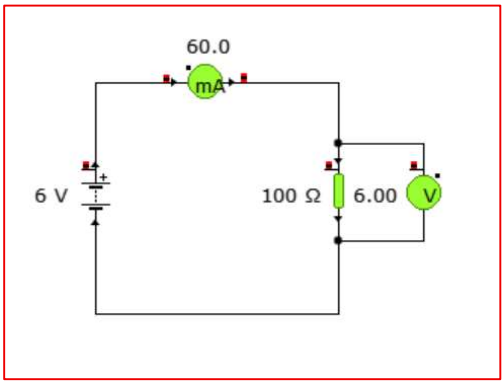
🧪 Task 1 – Verifying Ohm's Law

Learning Intentions

- To measure voltage and current in a simple circuit
- To verify Ohm's Law experimentally

Instructions

1. Build a simple circuit:
    - DC power supply (set to 6V)
    - 100Ω resistor
    - Ammeter (series)
    - Voltmeter (across resistor)



2. Record:
    - Supply voltage 6V
    - Current 60mA = 60 / 1000 = 0.06A
    - Measured voltage across resistor 6V
3. Calculate expected current using:

$$I = \frac{V}{R}$$

I = V/ R = 60/100 = 0.06A

4. Compare calculated and simulated values.

Extension

Change the supply to:

- 3V
- 9V
- 12V

Record results in a table and comment on the pattern.

Voltage (V)	Resistance (Ω)	Current (A)
3	100	0.03
9	100	0.09
12	100	0.12

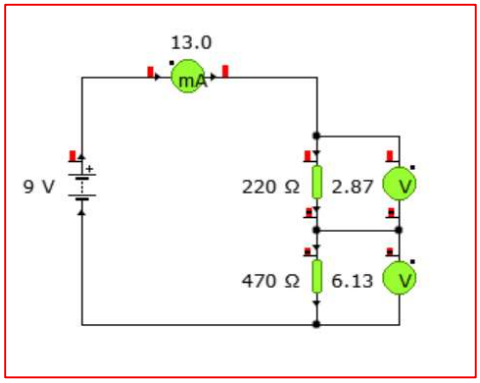
🔌 Task 2 – Series Circuits Investigation

Learning Intentions

- To investigate voltage sharing in series circuits

Build:

- 9V supply
- 220Ω resistor
- 470Ω resistor
- Ammeter in series
- Voltmeters across each resistor



Pupils must:

1. Measure total current. 13mA = 0.013A
2. Measure voltage across each resistor. $V_{220\Omega}$ = 2.87V, $V_{470\Omega}$ = 6.13V
3. Confirm:
    - Voltage adds to supply voltage. 2.87 + 6.13 = $9V_{Total}$
    - Current is the same everywhere. 0.013A = 13mA
4. Compare results with calculated values and discuss any issues found.

    $V_{220\Omega}$ = IR = 0.013 x 220 = 2.86V, $V_{470\Omega}$ = IR = 0.013 x 470 = 6.11V

    2.86 + 6.11 = $8.97V_{Total}$

    Difference between calculated and measured values is dependant on using correct overall current value. In the example, the current value was rounded to 0.013A. If we calculate, I = V/R = 9 / (220 + 470) = 0.1304A = 13.04A.

    Using 0.1304A gives:

    $V_{220\Omega}$ = IR = 0.01304 x 220 = 2.87V, $V_{470\Omega}$ = IR = 0.01304 x 470 = 6.13V

    2.87 + 6.13 = $9V_{Total}$

Challenge

Swap resistor values and predict results before simulating.

Swapping resistors will make no difference to the overall current or the voltages across each resistor.

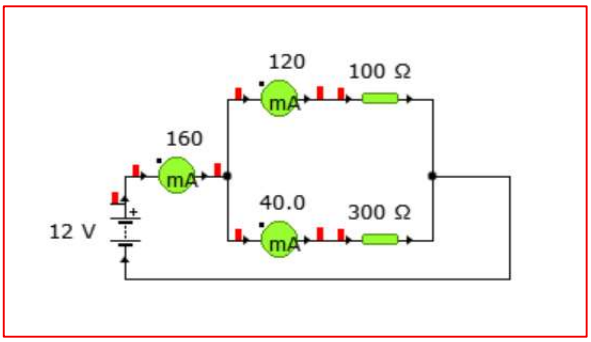
⚡ Task 3 – Parallel Circuits Investigation

Learning Intentions

- To explore current splitting in parallel circuits

Build:

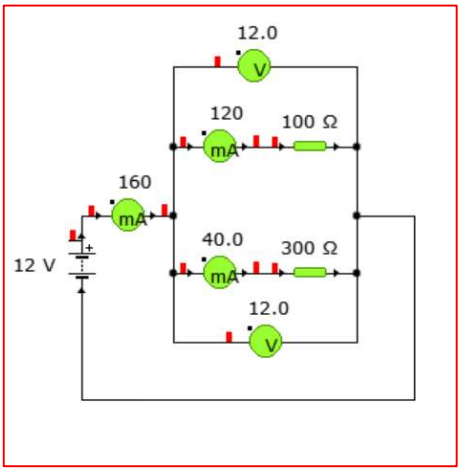
- 12V supply
- 100Ω resistor (both resistors placed in parallel)
- 300Ω resistor
- Ammeter in main line
- Ammeters in each branch



Pupils must:

1. Measure branch currents. $I_{100\Omega}$ = 120mA = 0.12A, $I_{300\Omega}$ = 40mA = 0.04A
2. Confirm:
    - Total current = sum of branch currents. I_{Total} = 0.12A + 0.04A = 0.16A
    - Voltage across each branch is equal.

      Each branch has 12V across it.



Extension

Add a third resistor and predict the effect on:

Total resistance

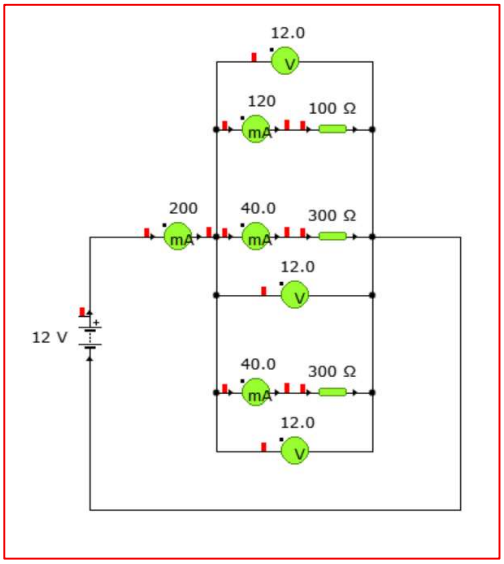
$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} = \frac{1}{100} + \frac{1}{300} + \frac{1}{300}$$

$$\frac{1}{R_T} = \frac{3}{300} + \frac{1}{300} + \frac{1}{300} = \frac{5}{300}$$

R_T = 60 Ω

- Total current

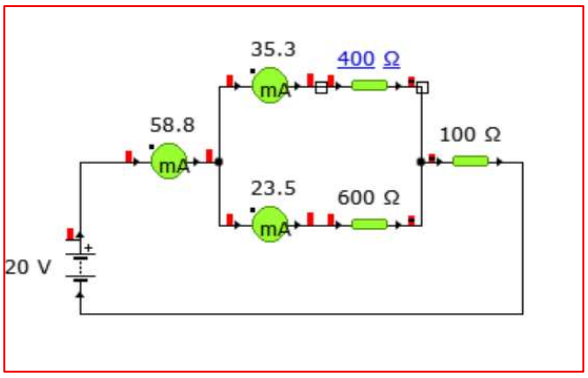
  R_T decreases and so I must increase.



🔀 Task 4 – Combined Circuit Challenge

Build:

- 100Ω resistor in series
- 400Ω and 600Ω resistors in parallel
- 20V supply



Tasks

1. Calculate total resistance before building.

   $$R_{Parallel} = \frac{R_1 \times R_2}{R_1 + R_2}$$

   $$R_{Parallel} = \frac{400 \times 600}{400 + 600}$$

   $R_{Parallel}$ = 240 Ω

   $R_{Total} = R_{Parallel} + 100$

   R_{Total} = 240 + 100 = 340Ω

2. Build circuit in Yenka.
3. Measure:
   - Total current 58.8mA = 0.0588A
   - Voltage across parallel section $V_{Parallel}$ = 14.1V
4. Compare calculated vs simulated results.

   Calculated: I = V/ R = 20 / 340 = 0.058824 = 0.0588A

   $V_{Parallel}$ = 0.0588 x 240 = 14.117647 = 14.1V

Reflection Question

Why is the current through the 400Ω resistor greater than through the 600Ω resistor?

Because the branch with the smaller resistance will allow more current through.

In parallel circuits, Current divides inversely proportional to resistance. Therefore,

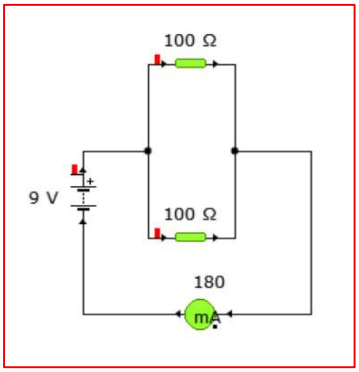
Lower resistance → higher current

Higher resistance → lower current

🛠 Task 5 – Design Challenge

Scenario

A student needs 50Ω but only has two 100Ω resistors.



Task

1. Use Yenka to create a 50Ω equivalent resistance.
2. Prove using measurement tools.

   R = V/ I = 9 / 0.180 = 50Ω
3. Screenshot and annotate their circuit.
4. Explain why the solution works.

   In parallel, both resistors have the same voltage across them.

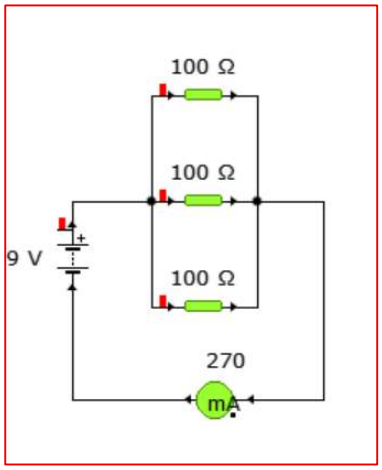
   The total current splits between the two branches.

   Because there are now two paths for current, the circuit allows twice as much current to flow compared to a single 100Ω resistor. This is the same as halving the resistance of one 100Ω resistor.

Extension

Create:

- 25Ω using three 100Ω resistors. The lowest resistance you can get using these resistors is 33.3 Ω. You **cannot** make 25 Ω with just three 100 Ω resistors.



- 75Ω using three 100Ω resistors.

  With three identical 100 Ω resistors, the only possible combinations give:

  300 Ω (all series)

  150 Ω (two parallel, then series)

  66.7 Ω (two series, then parallel)

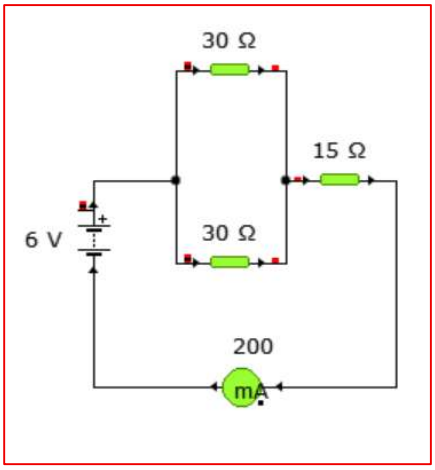
  33.3 Ω (all parallel)

  You cannot make 75 Ω exactly using only three 100 Ω resistors.

📊 Task 6 – Practical Data Task (Assessment Style)

Pupils must:

1. Build a circuit with:
    - 6V supply
    - 15Ω resistor in series
    - Two 30Ω resistors in parallel

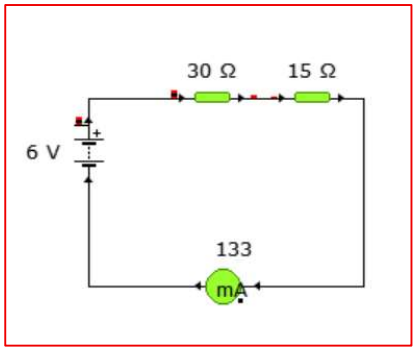


2. Complete a results table:

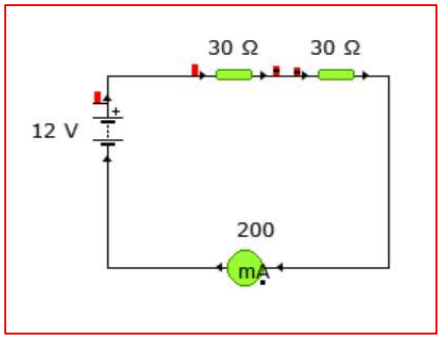
	Calculated	Simulated
Total Resistance (Ω)	30	-
Circuit Current (A)	0.2A	0.2A
Voltage across parallel section (V)	3V	3V
Current in one 30Ω resistor (A)	0.1A	0.1A

🧠 Higher Thinking Extensions

- Investigate what happens when a resistor in parallel is removed. Overall resistance increases and so overall current decreases.



- Investigate what happens when resistance increases in series. Overall resistance increases and so overall current decreases.
- Create a circuit that draws exactly 0.2A from a 12V supply.



- Build a circuit that divides voltage equally. A circuit with two identical resistors in series.

